Lab 4 — Final Submission Evidence (Answer Parts 1–9)

Student: Thanakorn Soison — 67070507205 — GitHub: @cottonlnwza **Repository:** cottonlnwza/toktickit **Final product release:** independently approved PR #83 merged `lab4-staging` to `main` at commit `64dd4f012af2f99b6a4c1f460599beeb9f712821` on 2026-10-31. The verification below was rerun from that exact post-merge commit. **Evidence status:** Product release and post-merge verification completed. The separate document/evidence update containing this report is subject to peer review; Issue #70/Kanban closure must be checked against actual GitHub status, not assumed.

Final main release verified

Merge SHA: `64dd4f012af2f99b6a4c1f460599beeb9f712821`

Peer review: PR #83 — APPROVED and peer merged

Test environment: local post-merge tests; isolated PostgreSQL test database; not Hosted CI.

Server	Client	Lab 4 E2E	Lab 3 E2E
243/243	116/116	33/33	7/7

Contents: Answer Part 1–9, source and review links, test evidence, nine real browser screenshots with readable mobile details.

Product release and post-merge tests are complete. The final evidence-update PR and Issue #70 closure remain subject to separate peer review.

Answer Part 1

Git Use with Engineering Workflow (10 points). The existing GitHub Project #3 implements Backlog → Specified → Started → PR Review → Fixing (as needed) → PR Review → Done. Feature PRs #71, #72, #73, #74, #75, #78, #79, #80, and #81 were independently approved and peer-merged to `lab4-staging`. The remaining release candidate evidence PR #82 was approved and peer-merged to `lab4-staging` as `629ea5b`; final release PR #83 was approved by `Tanaboonnnnn` and peer-merged into `main` as `64dd4f0`. This proves the full feature → staging → main lineage.

- Rendered peer-review record with reviewer identity, actual review comments, responses and approvals. Includes only genuine Lab 4 reciprocal reviews of another student's PRs.
- Repository structure, README, and .gitignore.
- **Release verified:** PR #83 and its review/merge SHA are recorded in `reviewer.md`. Final Issue #70 Done status and board evidence must be checked after the documentation/evidence update is peer-merged.

Answer Part 2

Spec DD (5 points). Rendered Sprint 4 specification describes 23 FRs, business rules, Actions Taken parent-child accountability, four-state Action lifecycle, eight Ticket statuses, server-enforced resolution gate, role dashboard calculations, authorization, optimistic revisions, preservation migration, idempotent seed and Product Definition of Done. API contract and UI contract supply the executable detail. The specification was peer-reviewed and merged in PR #71 **before** the implementation PRs #72 onward; see PR chronology.

The resolution gate requires at least one `COMPLETED` Action in the **current** `workflowCycle` and no `PLANNED`/`IN_PROGRESS` Actions. The Ticket primary owner is distinct from each Action assignee, and only the current assignee may complete that Action. Reopening begins a new cycle; older Actions remain historical. These are approved Lab 4 contract rules, not inferred additions.

Answer Part 3

Test DD and Traceability (10 points). Rendered Lab 4 test plan, 53 scenario IDs and AC traceability links actual unit, API, UI, migration, performance and E2E source files with status. Server Lab 4 tests, client Lab 4 tests, and Playwright Lab 4 tests are browsable.

Final `main` local verification, 2026-10-31 (exact SHA `64dd4f0`): Server **243/243** tests across 36 files in separately executed non-overlapping groups (Labs 1–2: 42; Lab 3: 136; Lab 4: 55; Lab 3–4 migration/seed: 10), Client **116/116** across 27 files, Lab 4 Playwright **33/33** (desktop/tablet/mobile 11 each), Lab 3 Playwright regression **7/7**, and both production builds passed. Tests with destructive schema setup used an isolated PostgreSQL `_test` database. The 243 server tests were **not** one combined test command. These are **post-release final-`main` runs**, not hosted CI.

Pre-release raw logs and the `artifacts/lab-04/verification/final-main/` files added by this evidence update preserve the executed commands for independent inspection. The final-main logs are labeled separately to prevent confusing them with earlier test runs.

Final `main` tests have been rerun after peer merge. Server totals come from four disjoint, serial test groups to isolate migration/seed database fixtures; the raw logs and exact SHA support this count.

Answer Part 4

AI Use with Reflection (5 points). Rendered AI-use and reflection document names ChatGPT and the connected coding bridge, selects **10 genuine Lab 4 key prompts** in the Lab 1 template, and includes a first-person reflection about scope discipline, TDD, correcting peer-review blockers, and separating AI suggestions from actual test/review evidence.

Answer Part 5

Working IT Staff Dashboard UI (5 points). The Staff/Admin Dashboard displays backend-calculated unassigned Tickets, my Tickets, my open Actions, Ticket status/IT-priority summaries, and recent or urgent Tickets. Metrics use current authenticated identity, current workflow cycle and active parent Ticket rules; they link to filtered Queue/Detail screens instead of becoming a separate Ticket-management interface. Selected counts were checked against direct PostgreSQL queries in the Issue #67 tests. Dashboard E2E covers metrics, drill-downs, role isolation and retryable failures.

Screenshots: Desktop · Tablet · Mobile. Loading/empty/forbidden/failure behavior has additional component/API/E2E checks documented in tests.md; not every state is separately pictured here.

Answer Part 6

Working Actions Taken UI (10 points). Actions Taken component and Lab 4 E2E flow demonstrate multiple Actions on a Ticket, immutable creation time, current/historical cycles, list/create/edit/reassign, `PLANNED` → `IN_PROGRESS` → `COMPLETED` or `CANCELLED`, required Result/follow-up note, invalid/inactive assignee rejection, Requester read-only restriction and protected API writes. Completed/cancelled/historical Actions cannot be edited. Backend version checks reject stale updates without partial writes. Browser flows operate through authenticated UI/API with the seeded PostgreSQL test database.

Screenshots: Desktop · Tablet · Mobile. They show multiple Action cards and a real field-level validation state; separate negative/race scenarios are supported by API and E2E tests rather than fabricated screenshots.

Answer Part 7

Working Ticket Workflow (5 points). Approved state matrix in the specification covers `NEW`, `OPEN`, `IN_PROGRESS`, `WAITING_FOR_REQUESTER`, `RESOLVED`, `CLOSED`, `REOPENED`, and `CANCELLED`; workflow API tests check allowed/rejected transitions, completed-current-cycle resolution evidence, reopen isolation, requester advisory indication, Administrator parity, and stale revision errors. Ticket resolution E2E provides browser-level proof. Actions Taken audit tests verify append-only Action events and deterministic version ordering; role-appropriate visibility is covered by backend enforcement and requester tests.

Answer Part 8

Working Requester Dashboard and Final Regression UI (5 points). Requester dashboard numbers and recent/attention Tickets are limited to the authenticated Requester's Tickets; ownership is enforced by the backend rather than by filtering another user's data in the browser. Drill-downs lead to My Tickets/Detail with the documented filter intent. Requester dashboard API tests and cross-Requester E2E show isolation. Lab 3 browser regression **7/7** covers authentication/password change, My Tickets/Ticket Detail and Requester continuation, Staff Queue/Detail, role authorization, and Administrator management. Labs 1–3 server/client regression suites also passed as recorded in Part 3.

Screenshots: Desktop · Tablet · Mobile.

Answer Part 9

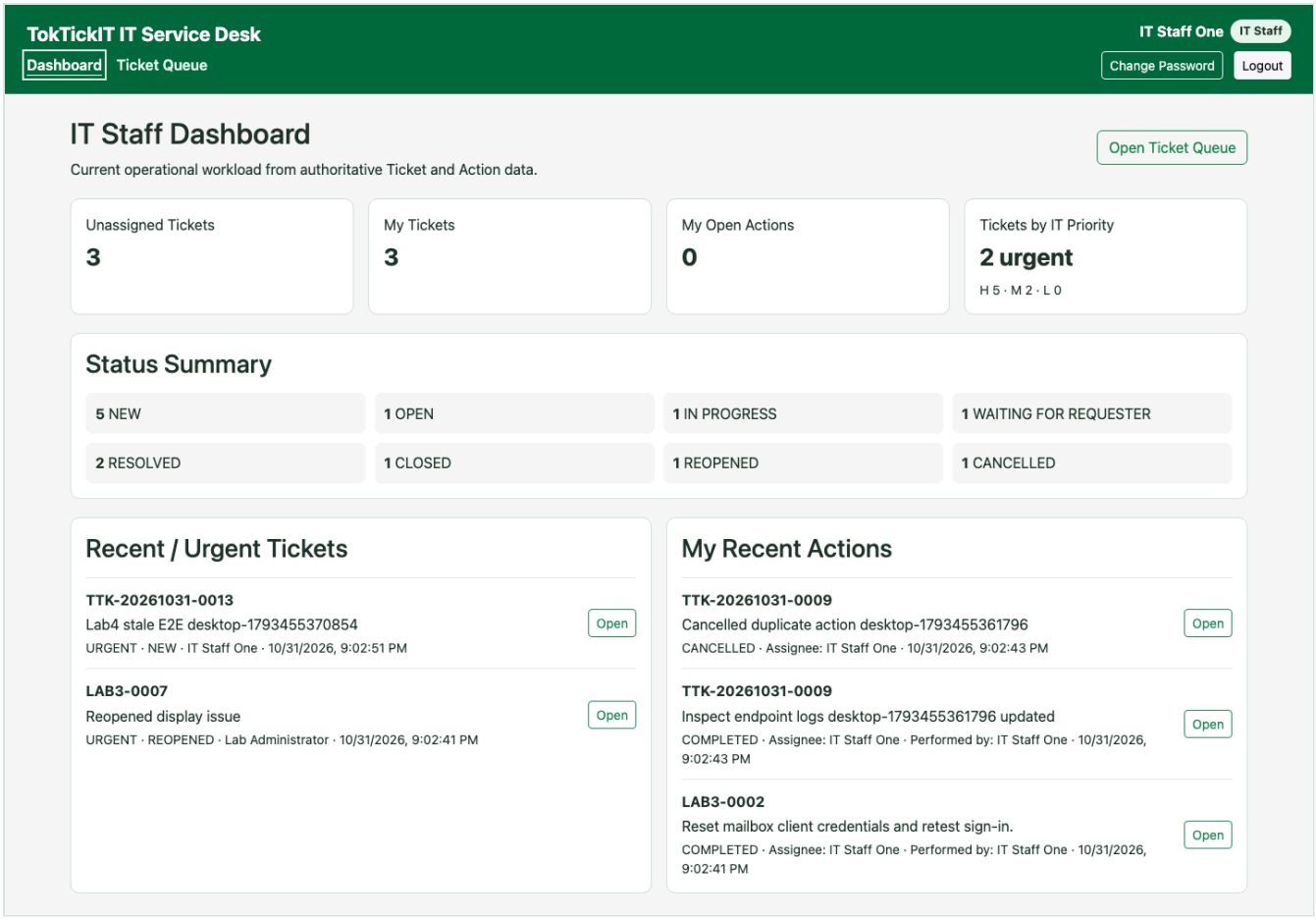
Zen Green UI, Responsive, Accessibility, and Final Polish (5 points). Rendered Lab 4 UI specification and completed visual/accessibility checklist. Browser checks at **1440×900 desktop, 820×1180 tablet, 390×844 mobile** verify no page-level horizontal overflow, hidden required controls or material clipping/overlap on the major screens; visible keyboard focus, field-associated validation, distinct read-only values and non-color status cues are also tested. Visual-hardening browser tests captured all nine real screenshots below.

Major screen	Desktop 1440×900	Tablet 820×1180	Mobile 390×844
IT Staff Dashboard	Image	Image	Image
Requester Dashboard	Image	Image	Image
Actions Taken	Image	Image	Image

Final audit note: Release PR #83 was independently approved and peer-merged; final-main unit, integration, E2E, and build results were actually run and recorded. Issue #70 should be moved to Done only when the final evidence-update PR is independently approved/merged and the final GitHub Project board is verified. The evidence PDF includes authentic screen captures and does not assert a future review/board event as already completed.

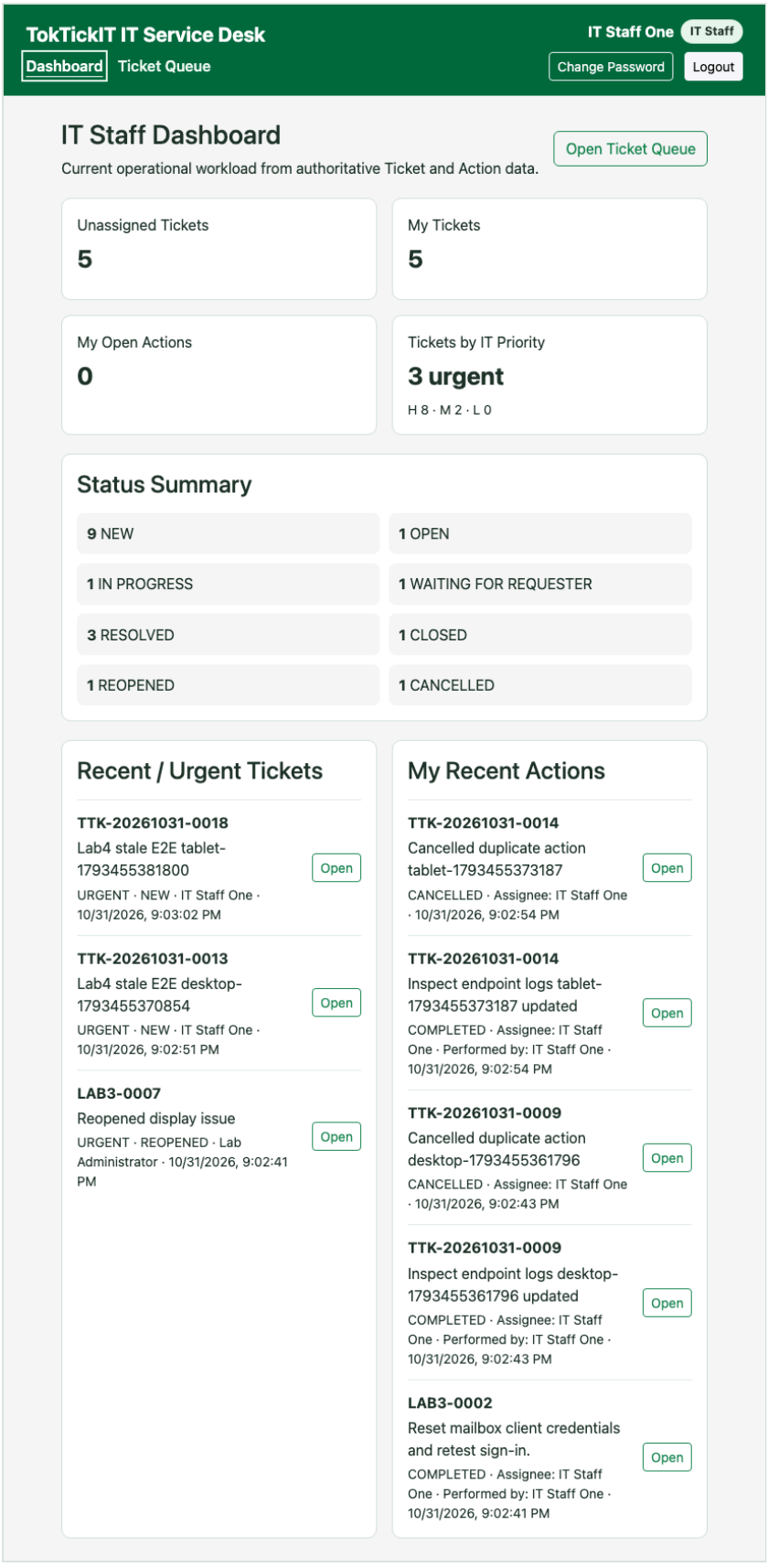
Answer Part 9 — Screenshot Evidence

Staff Dashboard / Desktop — actual final-main Playwright capture



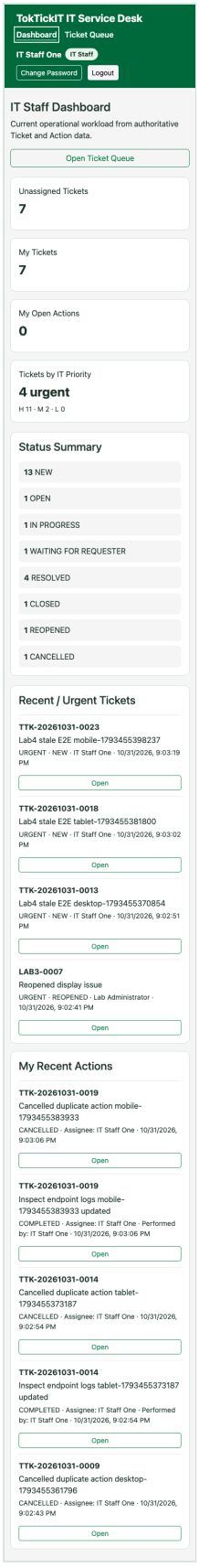
Answer Part 9 — Screenshot Evidence

Staff Dashboard / Tablet — actual final-main Playwright capture



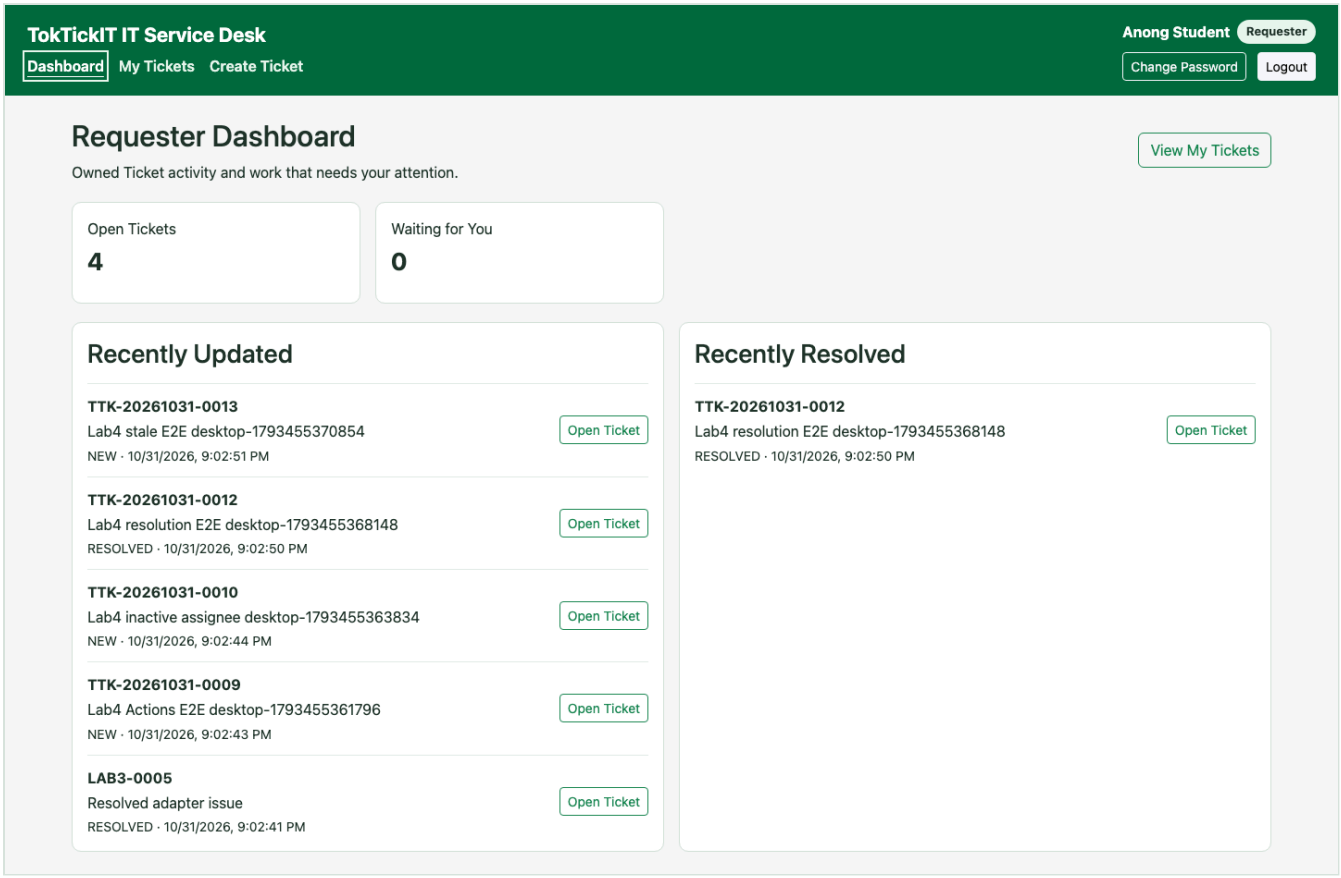
Answer Part 9 — Screenshot Evidence

Staff Dashboard / Mobile — actual final-main Playwright capture



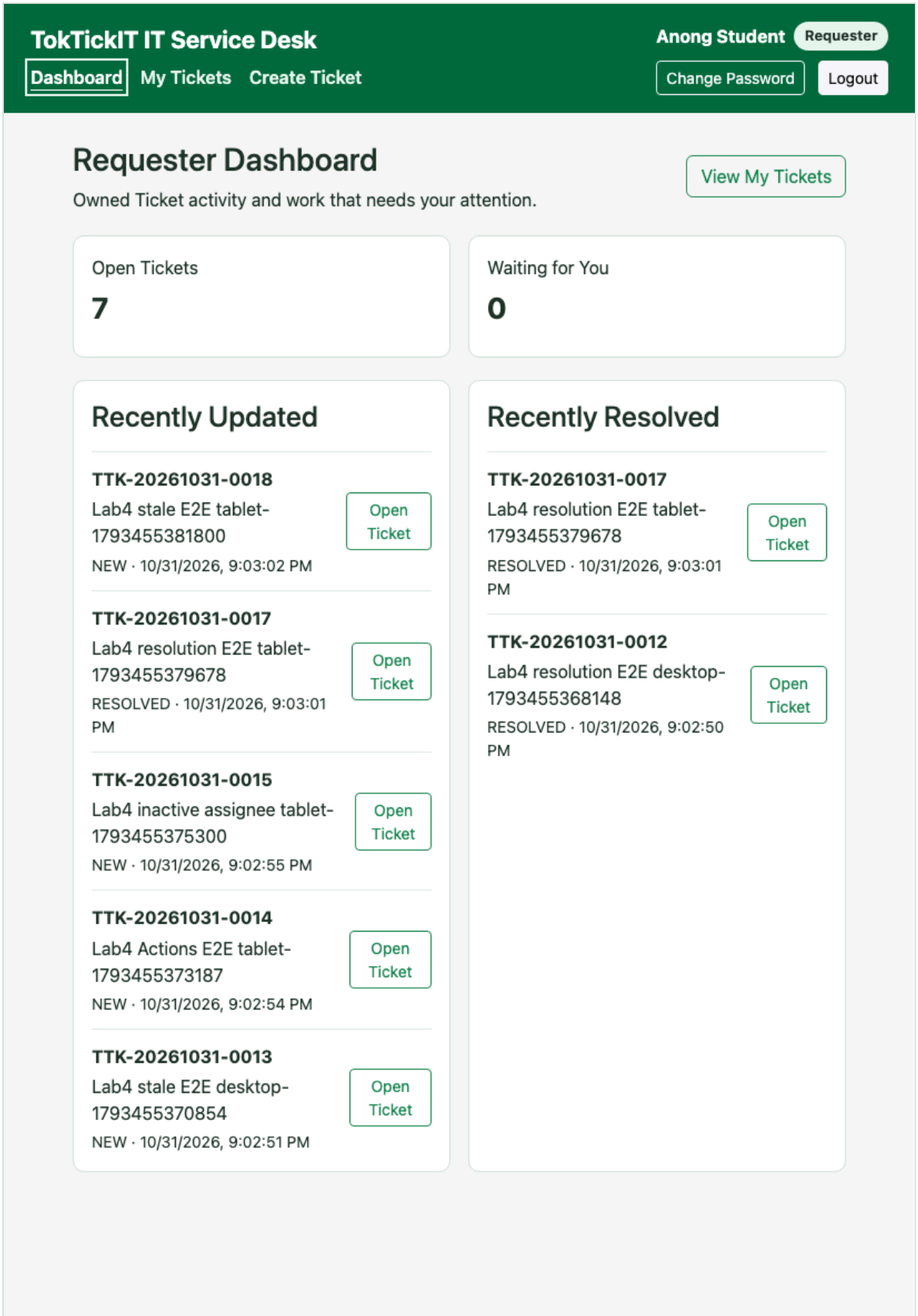
Answer Part 9 — Screenshot Evidence

Requester Dashboard / Desktop — actual final-main Playwright capture



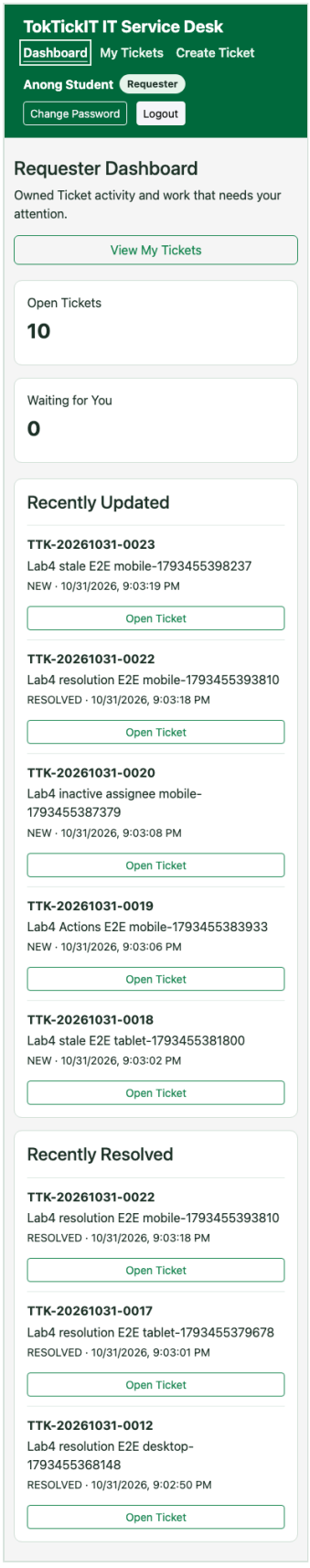
Answer Part 9 — Screenshot Evidence

Requester Dashboard / Tablet — actual final-main Playwright capture



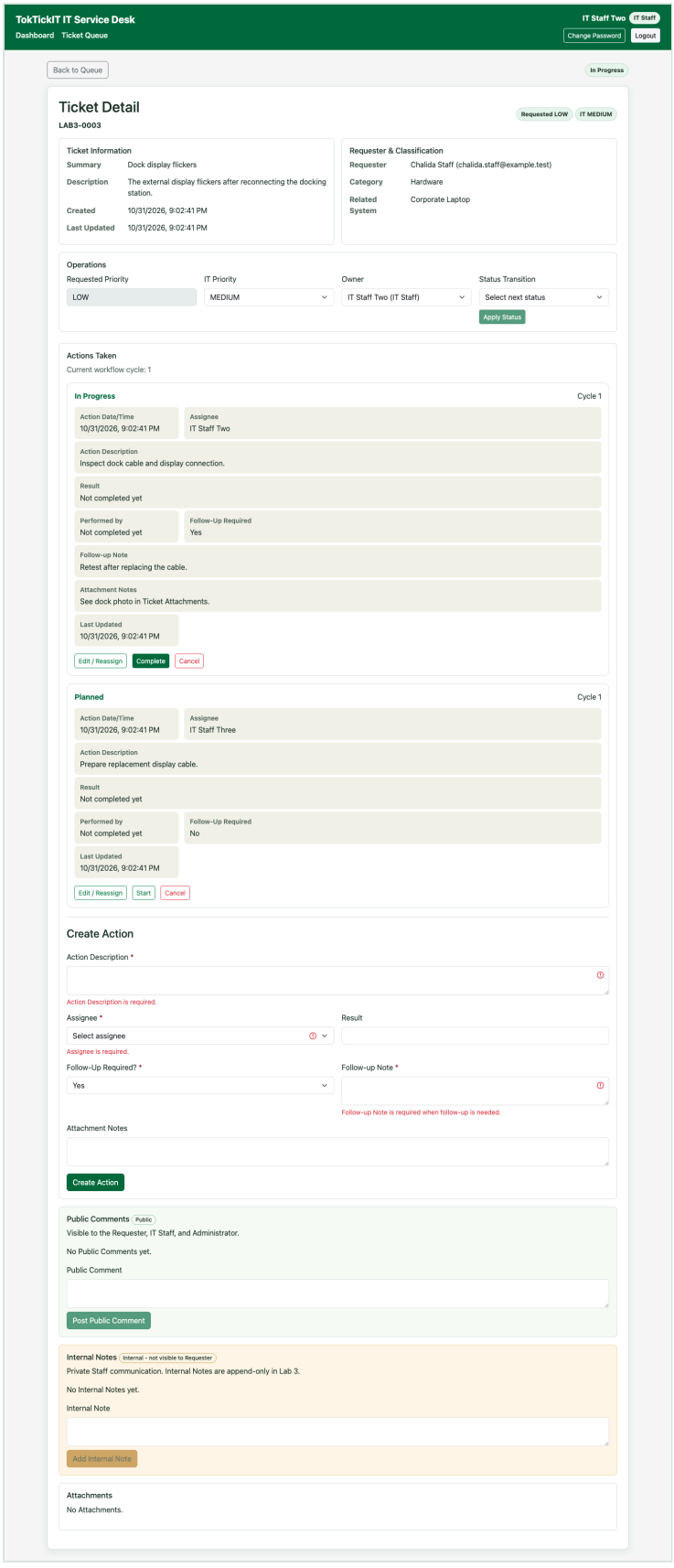
Answer Part 9 — Screenshot Evidence

Requester Dashboard / Mobile — actual final-main Playwright capture



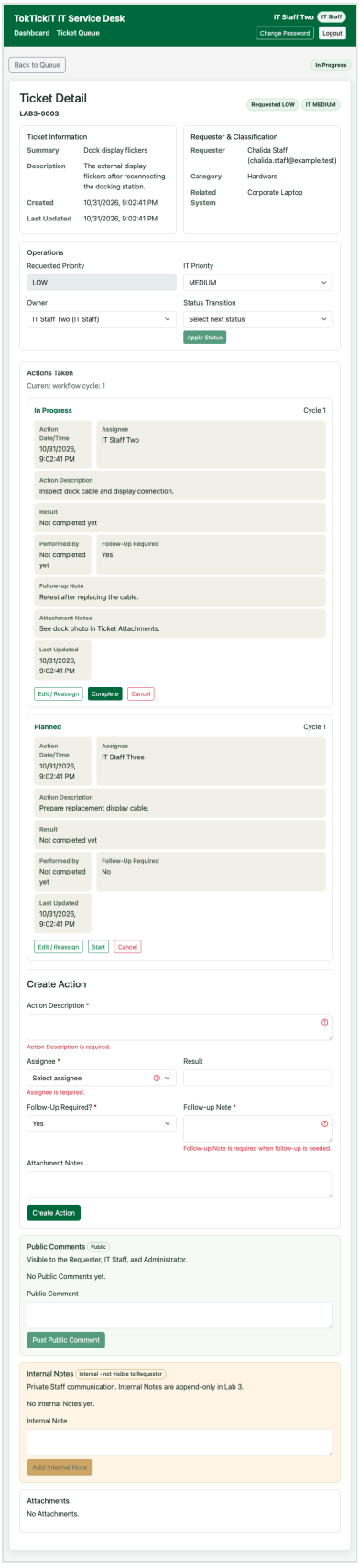
Answer Part 9 — Screenshot Evidence

Actions Taken / Desktop — actual final-main Playwright capture



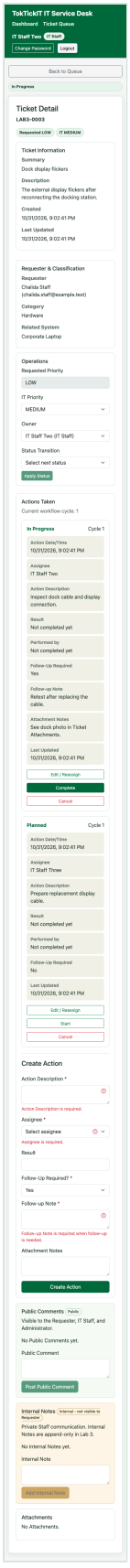
Answer Part 9 — Screenshot Evidence

Actions Taken / Tablet — actual final-main Playwright capture



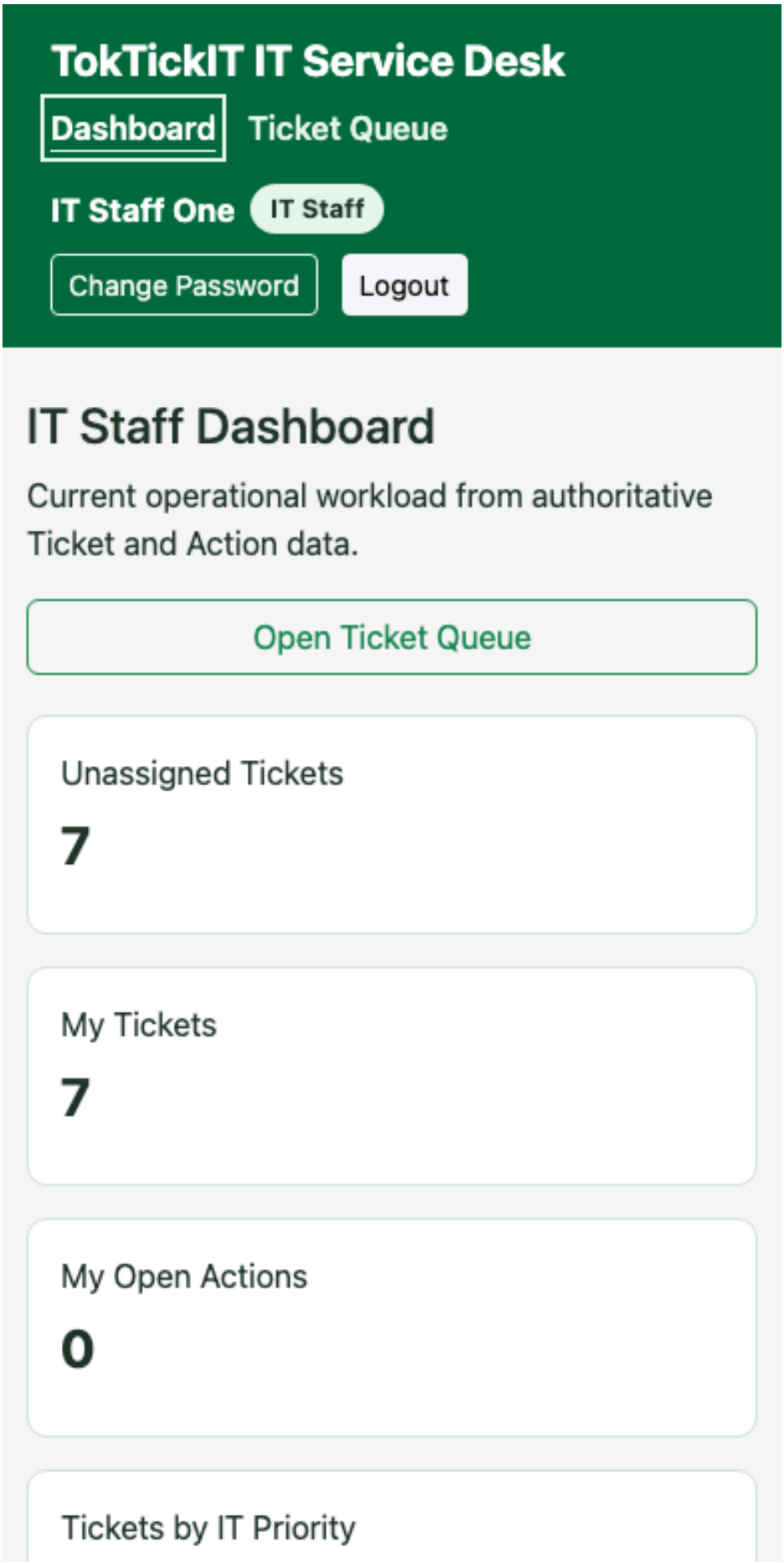
Answer Part 9 — Screenshot Evidence

Actions Taken / Mobile — actual final-main Playwright capture



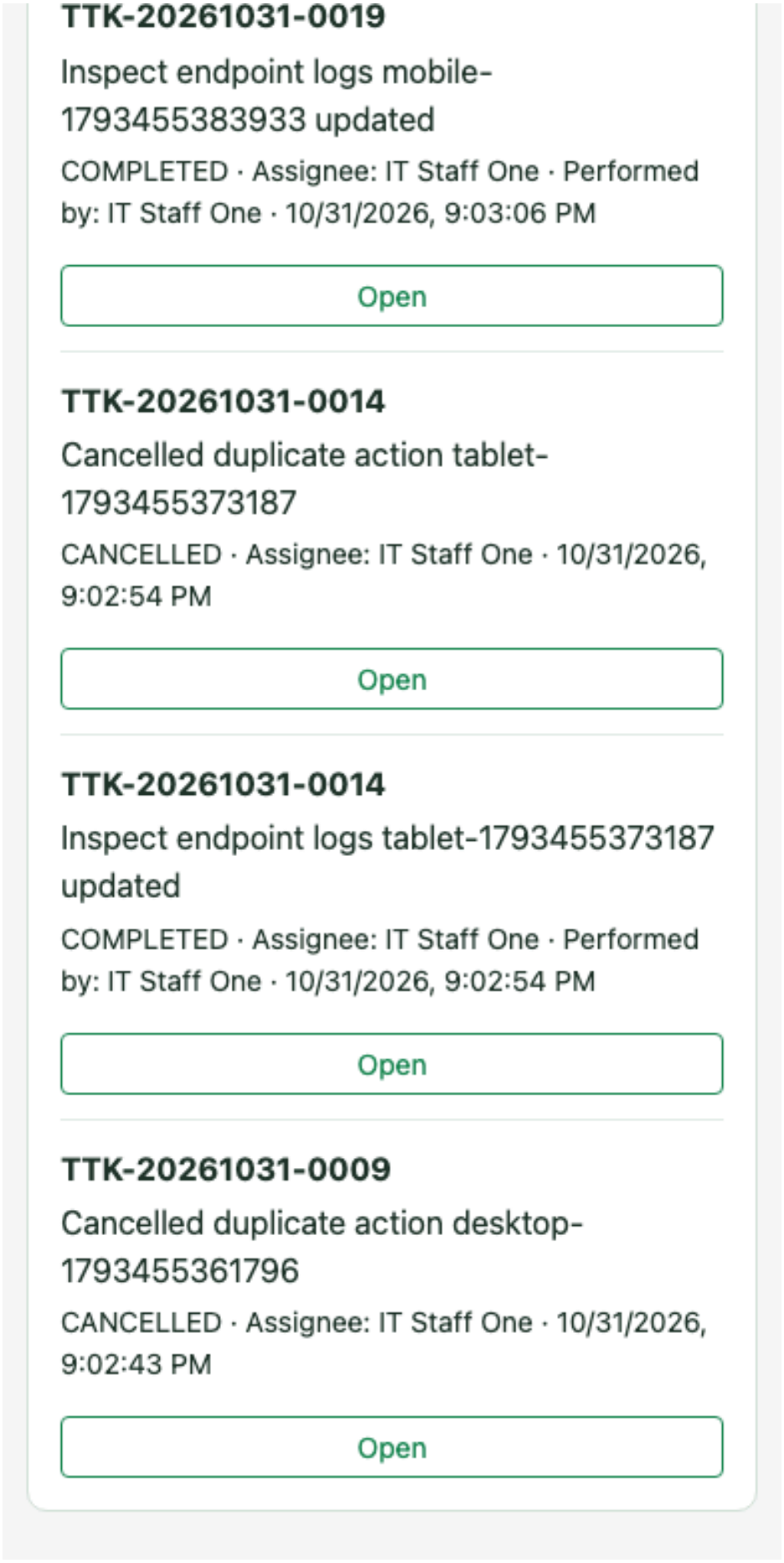
Answer Part 9 · Mobile screenshot detail

Staff Dashboard / original rows 0-780



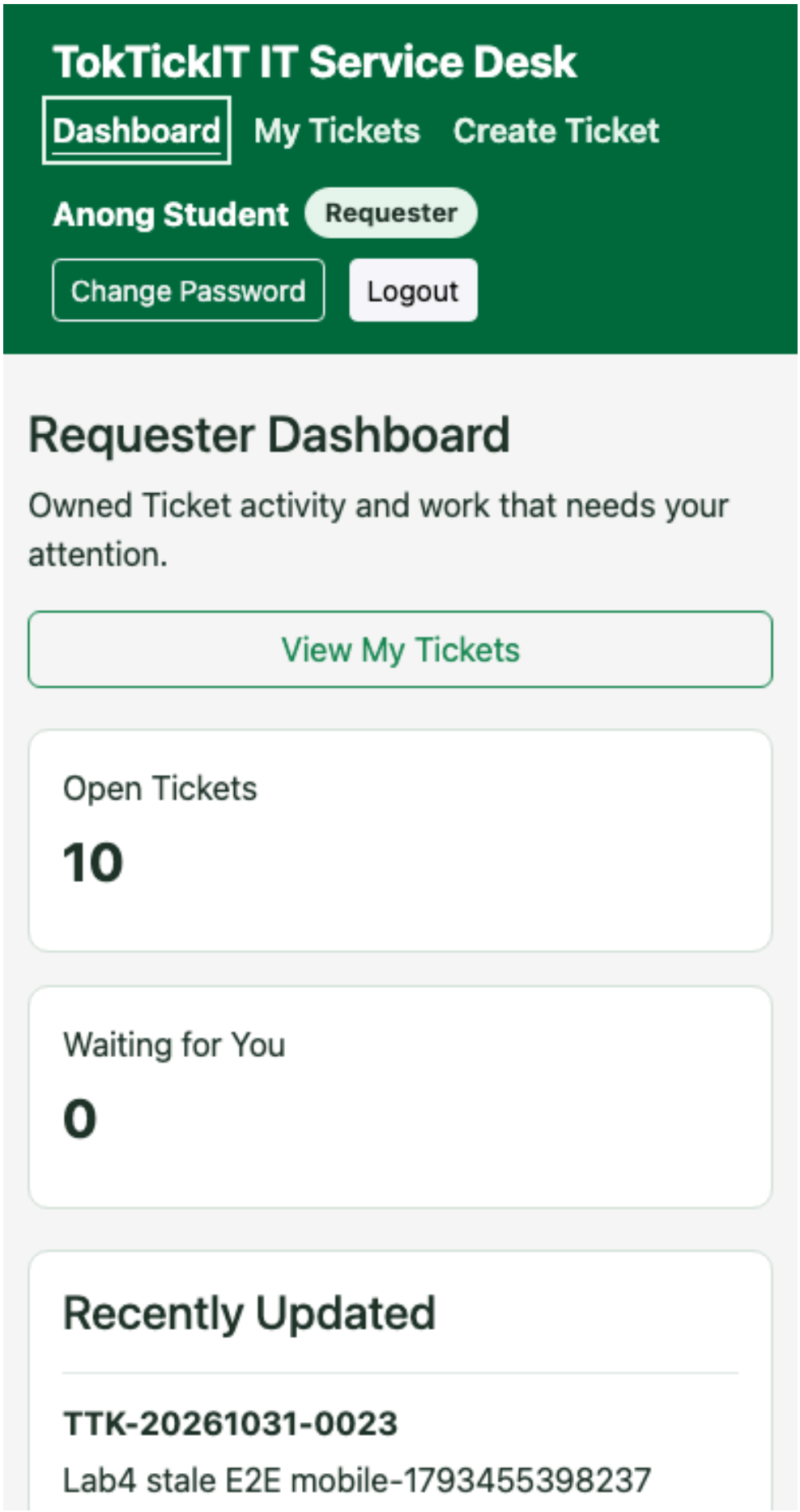
Answer Part 9 · Mobile screenshot detail

Staff Dashboard / original rows 2432-3212



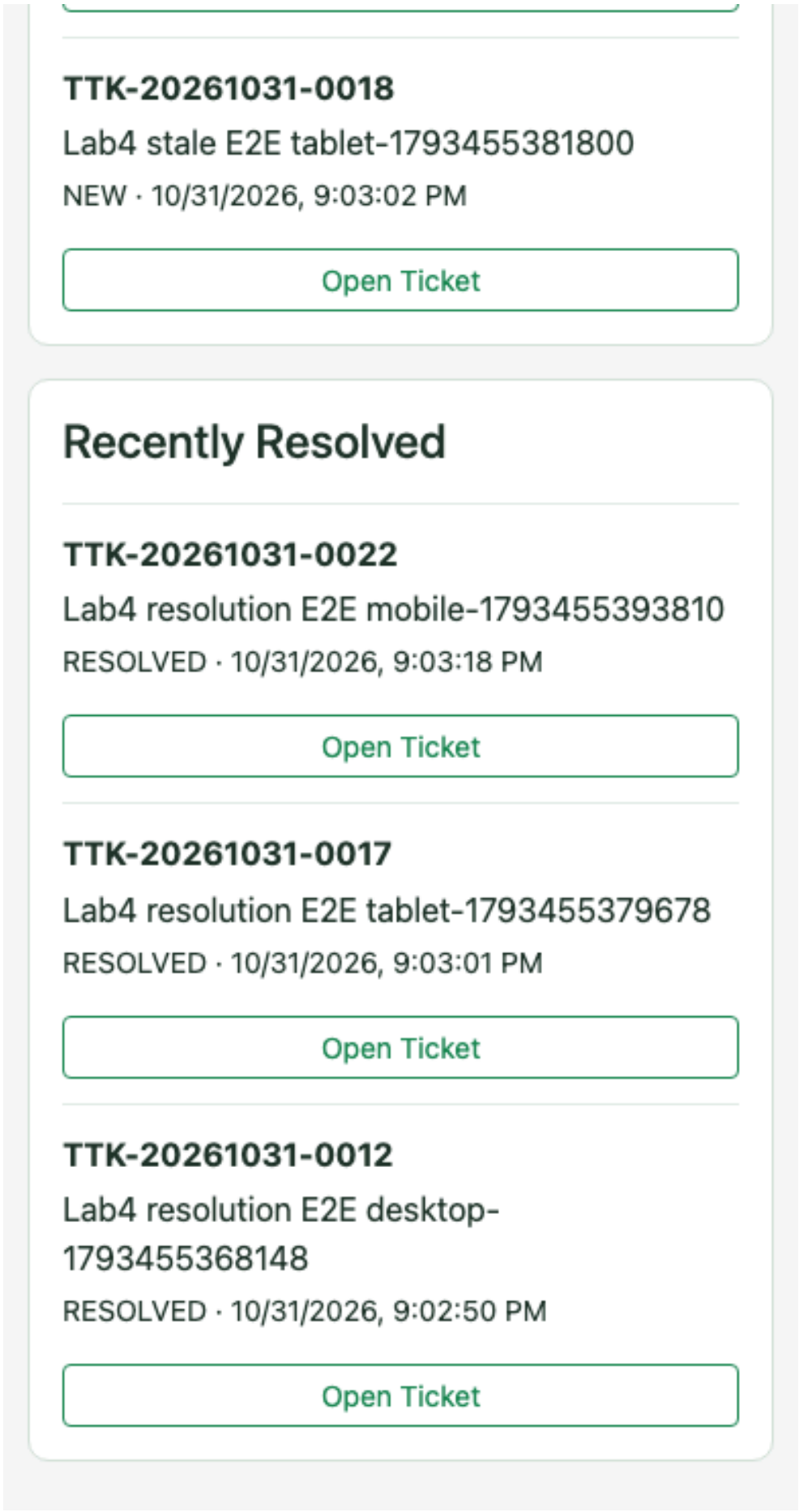
Answer Part 9 · Mobile screenshot detail

Requester Dashboard / original rows 0-740



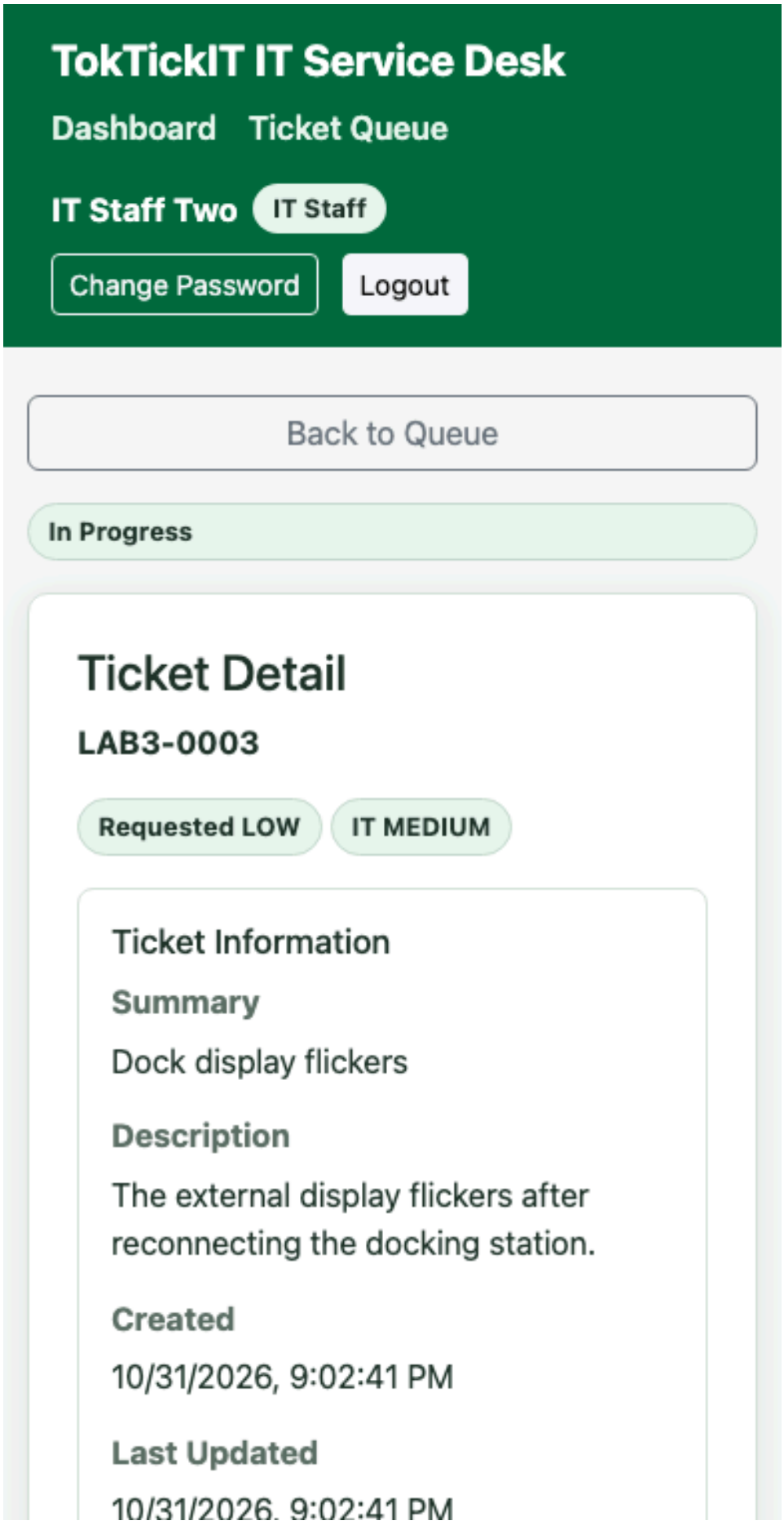
Answer Part 9 · Mobile screenshot detail

Requester Dashboard / original rows 1270-2010



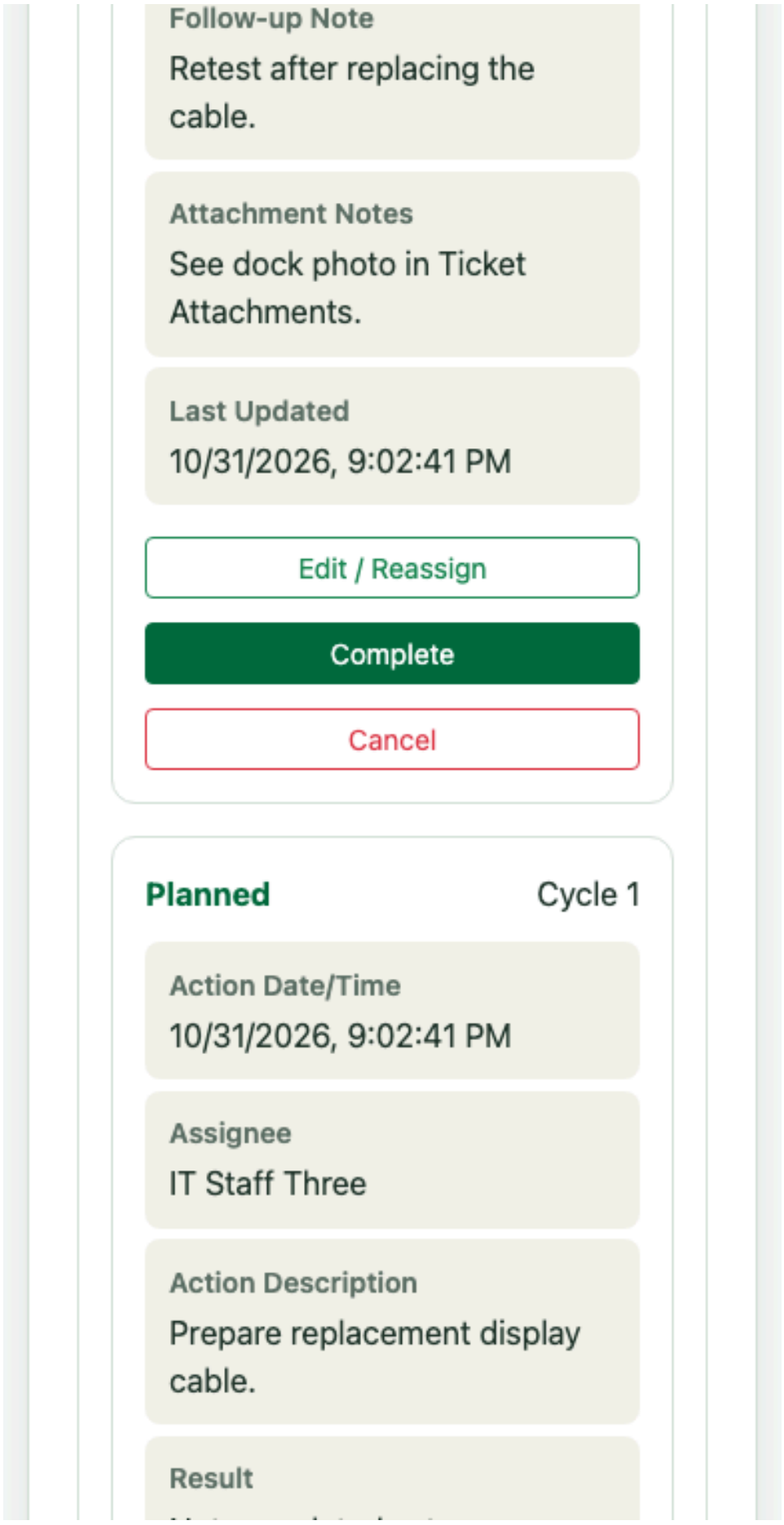
Answer Part 9 · Mobile screenshot detail

Actions Taken / original rows 0-760



Answer Part 9 · Mobile screenshot detail

Actions Taken / original rows 2200-2960



Answer Part 9 · Mobile screenshot detail

Actions Taken / original rows 4258-5018

